THE MECHANISTIC FIXED-EFFECTS TOPIC MODEL: RECOVERING CORPORATE CULTURE FROM EMPLOYEE REVIEWS

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How can employee reviews reveal comparable dimensions of corporate culture when workers in different industries and review settings describe similar experiences differently? We study this problem using over 900,000 pro and con documents from Glassdoor reviews of S&P 1500 firms across 20 industries. We introduce the mechanistic Fixed-Effects Topic Model (mFETM), which separates a shared topic component from environment-specific deviations. Topics are distributions over interpretable semantic features extracted from a language model using sparse autoencoders, retaining contextual information beyond word counts. Relative to a model without fixed effects, mFETM improves held-out document perplexity and reduces the association of estimated topic shares with review polarity and industry. Firm-level analyses relate these topics to inclusion in Fortune's 100 Best Companies to Work For and examine how the association changes when environment-associated topics are removed. Finally, we use the estimated fixed effects to steer language generation and assess the resulting changes in review polarity using illustrative generations and held-out likelihood exercises. These results support a framework for measuring shared workplace themes while accounting for systematic differences in how reviews express them.

1. Introduction. Social scientists use topic models to turn large text corpora into interpretable measures of latent constructs (Grimmer and Stewart, 2013; Gentzkow, Kelly and Taddy, 2019). Applications span political agendas, legal briefs, managerial behavior, and central-bank deliberations (Quinn et al., 2010; Sim, Routledge and Smith, 2015; Hansen, McMahon and Prat, 2018; Bandiera et al., 2020). Topic models can recover ideologies, narratives, and organizational practices that are otherwise difficult to observe at scale.

Corporate culture is a natural application. Persistent differences in how organizations operate may help explain performance gaps even after observable inputs are held fixed (Schein, 1985; Syverson, 2004; Kotter and Heskett, 2011; Gibbons and Henderson, 2013). Although a long literature classifies these differences into recurring cultural types (Handy, 1978; Cameron and Quinn, 2011), culture remains difficult to measure at scale. Employee-review websites are particularly useful in this context, as they provide a firsthand accounts of how workers experience their firms (see, e.g., Cai, Prat and Yu, 2024). Turning those reviews into comparable measures of corporate culture, however, raises two problems for traditional topic models.

The first is environmental heterogeneity. In a corpus drawn from many sources, a common theme may be expressed differently across industries, firms, or document types. A worker describing rigid schedules may mention pick rates and timed breaks in a warehouse, but call

Keywords and phrases. Corporate culture, Topic models, Fixed effects, Sparse autoencoders, Text as data.

TABLE 1
Two Reviews Describing Rigid Work Schedules

Company	Glassdoor Review
Amazon	*"Ran like the military, very cutthroat environment. You are given rates you need to pick and pack that are very hard to make if you don't have the work in front of you. Thirty minute lunches and 15 minute breaks however you must walk about 5–10 minutes to get your lunch (if it's not stolen) so even that you don't look forward to."*
Aflac	*"Being on the phone all day is kind of like jail. Pre-scheduled breaks and lunch times. An adult should be able to choose when they eat lunch. Sometimes they scheduled your lunch after you'd only been at your shift for an hour or two. Then you are stuck at your desk the rest of the day. Getting time off can be a hassle. It's determined by a computer, not your supervisor. Very little to no advancement opportunities."*

queues and computer-assigned time off in an office. A standard topic model can interpret these differences in expression as differences in substance. We refer to this source-specific language as an environmental language fixed effect.

The second problem is representation. Traditional topic models (Blei, Ng and Jordan, 2003) treat each document as a bag of word counts. Under this representation, structure is captured by word co-occurrence, so potentially important semantic objects, like context and tone, are discarded. Corporate concepts often appear through combinations of phrases, metaphors, and evaluations whose meaning depends on context, which limits a bag-of-words representation for this task. Word-based topics are also difficult to interpret, as their meaning must be inferred from a ranked list of words.

The two reviews in Table 1 make both problems concrete. A warehouse associate at Amazon and a customer-service specialist at Aflac both complain about rigid schedules and tightly regulated routines. Yet the first describes rates, packing, and breaks, while the second describes phones, desks, and computer-assigned time off. Their common complaint is expressed through task-specific vocabulary, metaphor, and tone that simple word counts can only partially capture.

We develop a topic model that addresses both problems jointly, the mechanistic Fixed-Effects Topic Model (mFETM). The model has two components. First, mFETM places fixed effects inside the generative process: a global backbone captures the component of each topic shared across environments, while environment-specific deviations capture how it is expressed in a particular industry, review polarity (pros or cons), or other observed environment. Second, it replaces the bag-of-words representation with a bag of concepts. This representation is built from counts of automatically labeled semantic features extracted from a language model by a sparse autoencoder (SAE). A topic is therefore a distribution over readable concepts rather than words.

To predict language, a large language model develops internal representations of recurring meanings, such as schedule flexibility, supportive coworkers, or customer service. These meanings are mixed together across many coordinates and are therefore difficult to read directly. An SAE decomposes those dense internal representations into a sparse set of recurring features, meaning that for any passage only a small number are active. Passages that strongly activate the same feature help identify its common meaning, allowing the feature to be labeled and used as an interpretable concept.

We estimate mFETM on a large corpus of Glassdoor reviews and compare it with a mechanistic vanilla topic model, mVTM. The benchmark uses the same concept representation and the same number of topics but omits the fixed effects.[1] The comparison holds the data,

[1] Zheng et al. (2025) establish the advantage of the concept representation over bag-of-words for vanilla topic models; we instead study the contribution of fixed effects.

representation, topic count, and encoder architecture fixed while assessing the contribution of incorporating observed environments during topic estimation.

We report three sets of results. First, we test the property the fixed effects are designed to deliver: topics should vary less across observed environments and reveal less about them. Between pros and cons reviews, mFETM's average topic-share spread is less than half mVTM's. The spread is also lower across industries, although to a smaller degree. We also test how well a review's topic shares predict its polarity and industry. Topic shares from mVTM explain 55.4% of the variation in whether a review is a pro or a con, compared with 19.7% for mFETM. They also distinguish industries significantly more accurately in all 28 pairwise sector comparisons we consider. Thus, mFETM's topic shares contain less information about the observed environments.

Second, we relate the estimated topics to inclusion on Fortune's 100 Best Companies to Work For, which we use as an external benchmark of employer quality. The estimated relationships show that mFETM captures a broader, more interpretable range of workplace dimensions, spanning compensation, work-life balance, coworker relations, management, restructuring, and workplace amenities. mVTM also identifies general workplace attributes, but mixes them with environment-specific predictors. In other words, it therefore provides a less complete map of the general workplace dimensions that distinguish top employers.

We then examine how much each model's fit depends on topics associated with industry and review polarity. We rank topics by these associations, progressively exclude the topic shares of the most strongly associated topics from the Fortune model, and re-estimate it using the remaining topic shares. As these predictors are removed, mFETM retains substantially more of its original fit than mVTM. This implies that its ability to distinguish top employers depends less on topics specifically associated with industry and review polarity.

Finally, combining the SAE representation with the fixed-effects model allows us to map topics and environment shifts to directions in the language model's representation space and steer generation along them. For a given topic, we steer the model along an estimated environment direction and examine changes in industry vocabulary and evaluative framing, holding the topic index and prompt fixed. This exercise allows us to manipulate the estimated fixed effects during generation, in addition to using them as statistical controls.

We return to the Amazon and Aflac reviews to illustrate the modeling choices. Supplementary Table A1 reports the bag-of-words estimates that originally motivated the project. A vanilla topic model assigns the reviews to different topics reflecting industry and polarity, despite their common complaint. Adding fixed effects gives both reviews the same leading topic by absorbing some industry- and polarity-specific language. Yet the shared topic is still represented by a list of words (workplace, lunch, food), rather than an explicit concept capturing the common concern: rigid schedules and tightly regulated routines.

The bag of concepts addresses this second limitation. Building on Zheng et al. (2025), we represent each document by the labeled features it activates inside a language model. Topics become distributions over those concepts. Supplementary Table A2 presents word- and concept-based topics side by side to show how concept labels make topics easier to interpret.

After estimation, a language model assigns each topic a short name using its concept list and the 20 reviews in which that topic has the highest estimated share. The name describes the workplace theme shared by the concepts and reviews (Section 4.1 and Supplementary Section D).

Table 2 shows how mFETM captures a shared concern despite differences in workplace vocabulary. Attendance and time-off policies accounts for 12.4% of the Amazon review and 27.8% of the Aflac review, linking their complaints about rigid breaks and limited control over time off. The topic shares also capture differences in emphasis: warehouse operations is

TABLE 2
Top mFETM Topics for the Two Reviews

Topic (mFETM)	Share
Amazon	
Warehouse Operations and Equipment Handling	0.257
Attendance and time-off policies	0.124
Wages and Working Hours	0.116
Aflac	
Attendance and time-off policies	0.278
Warehouse Operations and Equipment Handling	0.172
Variable work schedules and hours	0.109

Notes. Each review's top three mFETM topics from the estimated model. Share is the review's estimated topic share. Two topics appear in both top-three sets, but the leading topic differs.

the leading topic for Amazon, while attendance and time-off policies is the leading topic for Aflac.

Together, these steps turn employee reviews into interpretable measures of corporate culture. Fixed effects account for industry vocabulary and review polarity, while the concept representation and topic names express the results as recognizable workplace themes. This allows mFETM to connect common workplace experiences across different organizational settings, even when workers describe them in different terms.

The paper contributes to three literatures. Studies of corporate culture use text to measure culture and relate it to firm outcomes (e.g., Guiso, Sapienza and Zingales, 2015; Li et al., 2021; Cai, Prat and Yu, 2024). We develop an unsupervised, interpretable measure that accounts for industry and review polarity. Topic models such as structural topic models, sparse additive models, and neural variants incorporate document covariates into the generative process (Roberts et al., 2014; Eisenstein, Ahmed and Xing, 2011; Card, Tan and Smith, 2018). mFETM estimates these effects over concepts rather than words. Finally, research on language-model representations identifies concept directions and recovers interpretable features using sparse autoencoders (Bricken et al., 2023; Zheng et al., 2025). We use these features to map estimated topic-specific environment shifts to directions that can steer language generation.

The rest of the paper proceeds as follows. Section 2 describes the Glassdoor data and the concept representation. Section 3 introduces mFETM, its estimation, and the mVTM benchmark. Section 4 presents the estimated topics and compares held-out fit. Section 5 measures the topics' association with industry and review polarity. Section 6 relates the topics to Fortune-list inclusion and tests how removing environment-associated topics affects fit. Section 7 uses the estimated fixed effects to steer language generation. Section 8 concludes.

2. Representing Text with Bags of Concepts. We begin by describing the data and the construction of the bag-of-concepts representation. We represent each document by counts of labeled semantic concepts rather than counts of words. This captures information that word counts miss and makes the estimated topics easier to interpret. Building on Zheng et al. (2025), we obtain both the concept vocabulary and the document-level counts from a pretrained sparse autoencoder.

2.1. *Glassdoor Data.* We analyze employee reviews from Glassdoor, an online platform on which current and former employees anonymously review their workplaces. The dataset covers S&P 1500 firms during 2008–2022 and has been used to study organizational capital

(Cai, Prat and Yu, 2024). Related studies use Glassdoor data to examine wages, employee perceptions, and stock returns (Karabarbounis and Pinto, 2018; Lee et al., 2021; Green et al., 2019). The platform's "give-to-get" policy encourages users to contribute reviews to access content and has been shown to reduce selection bias in ratings (Marinescu et al., 2021).

The reviews describe workplace experiences across firms and industries, making them well suited to testing whether mFETM can separate common cultural themes from context-specific language.

2.2. *Sparse Autoencoders.* Sparse autoencoders (SAEs) were developed to interpret large language models (Cunningham et al., 2023; Bricken et al., 2023). As a language model reads a review, it represents each token and its context with a high-dimensional activation vector a. Following the mechanistic-interpretability literature, we treat recurring semantic features as directions in that activation space (Bricken et al., 2023; Zheng et al., 2025). Many features share the same coordinates, making individual coordinates difficult to interpret (Elhage et al., 2022).

The SAE approximates a with a sparse, nonnegative combination of directions from a dictionary of h features:

$$r = \sigma(W_{\text{enc}}\, a + b_{\text{enc}}) \in \mathbb{R}^h_{\geq 0}, \qquad \hat{a} = W_{\text{dec}}^\top r + b_{\text{dec}}.$$

The encoder maps the residual-stream activation a to feature coefficients r, with one coefficient per dictionary direction. The SAE uses a JumpReLU function: each coefficient is retained if it exceeds a learned feature-specific threshold and is otherwise set to zero. The decoder reconstructs a by combining the rows of W_{dec} with weights r. The parameters are estimated to minimize reconstruction error while encouraging sparsity in r.

In practice, this fit–sparsity trade-off often produces directions aligned with individual concepts, such as mentions of long commutes or expressions of gratitude (Bricken et al., 2023). At each token position, only a small number of entries of r are positive. When one is positive, we say the corresponding concept fires.

SAE features are interpreted using passages in which they activate strongly. A language model summarizes the common content of these passages in a short description. Such descriptions can be evaluated by how well they predict feature activations on new text (Bills et al., 2023). We use the feature descriptions available through Neuronpedia and refer to each labeled feature as a concept. Table 3 lists the concepts that fire in the largest share of documents in our corpus.

We use a pretrained SAE and hold its parameters fixed throughout the analysis. Passing a review through the language model and SAE gives feature activations at each token position. For each retained feature, we count the token positions at which its activation exceeds a feature-specific threshold. These counts form the document's bag of concepts.

2.3. *Constructing the Bag of Concepts.* We next convert the raw reviews into the bag-of-concepts count matrix used by the topic models. We summarize the procedure here and report implementation details in Supplementary Section C.

We treat each review's "pros" and "cons" sections as separate documents carrying a polarity label. After the document-level filters, the corpus contains $906{,}938$ documents: $453{,}215$ pros and $453{,}723$ cons.[2]

Each document is processed by Gemma-2-9B, an open language model (Gemma Team, 2024). At each token, we take the internal representation a described in Section 2.2 and pass it

[2]The reviews are a 20% random sample of the corpus described in Section 2.1, restricted to the 20 two-digit NAICS industries with more than $1{,}000$ documents.

TABLE 3
Most Common Concepts in the Corpus

Concept	Share of documents
Elements related to personal connections and experiences in social contexts	74.0%
Elements related to shopping experiences and customer service	62.9%
Mentions of job performance and employment situations	54.7%
Aspects related to food quality and dining experiences, particularly emphasizing price, taste, and service	51.9%
References to family relationships and genealogical connections	48.9%
Various positive adjectives and expressions of approval	46.7%
Issues related to ethics and accountability in business practices	43.7%
Terms related to employee rights and leaves of absence under the Family Medical Leave Act (FMLA)	42.8%
Criticisms of food quality and service in restaurant reviews	41.1%
Positive descriptors related to quality and performance	40.7%

Notes. The ten concepts firing in the largest share of documents in the final thresholded count matrix, with their labels, after the pruning of Section 2.3.

through a pretrained Gemma Scope SAE (Lieberum et al., 2024). This identifies the concepts that fire and their activation strengths. We prune concepts that are too rare or common to be informative, along with those describing syntax rather than content, and count how often each retained concept exceeds its activation threshold:

$$c_{d,v} = \sum_{j} \mathbf{1}\{\alpha_v(a_{d,j}) > q_v\},$$

where the sum runs over the tokens of document d, $\alpha_v(a_{d,j})$ is concept v's activation strength at token j, and q_v is a concept-specific threshold. Supplementary Section C reports the SAE layer, pruning rules, and thresholds. The final vocabulary contains $V_{\text{kept}} = 4{,}735$ labeled concepts; the counts $c_{d,v}$ are the inputs to both topic models.

3. The Mechanistic Fixed-Effects Topic Model. The previous section turned each review into counts of interpretable concepts. Pooled topic estimation can still mix shared themes with environment-specific language. We now formalize this problem and specify a model that addresses it during estimation. Figure 1 summarizes the full pipeline.

3.1. *The Fixed-Effects Problem.* Social scientists often use topic shares as downstream measures, comparing what is discussed across firms, industries, or time. Such comparisons require topics to have a common meaning across environments, which pooled topic models need not preserve.

Consider a workplace-conditions theme. Industry vocabulary and review polarity may split it into sector-specific topics or mix it with expressions of dissatisfaction. Differences in topic shares across sectors may then reflect differences in language rather than workplace conditions. The resulting measurement error is systematic because it depends on the environment. At the sane time, adding industry or polarity controls in a downstream regression does not resolve the underlying problem, as the topic shares themselves may already measure different things in different environments.

mFETM accounts for environment-specific language during topic estimation. Each topic has a shared component and environment-specific deviations that describe how the theme is expressed. This structure allows the model to capture differences in expression without assigning them to separate topics, making topic shares more comparable across environments.

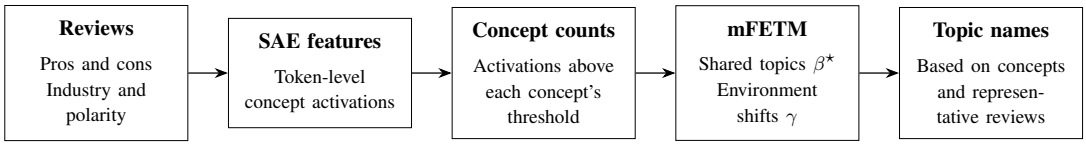


FIG 1. *From employee reviews to corporate-culture measures. A pretrained language model and SAE provide token-level concept activations, which are thresholded and counted within each document. mFETM estimates shared topics and environment-specific shifts from these counts. A language model names each topic using its concepts and representative reviews. The mVTM benchmark sets* $\gamma = 0$.

3.2. *Model Specification.* The model takes as input the corpus $\mathcal{D} = \{(\mathbf{c}_i, \mathbf{x}_i)\}_{i=1}^n$ constructed in Section 2. Each document has a concept-count vector $\mathbf{c}_i \in \mathbb{Z}_{\geq 0}^V$, where V is the number of concepts, and an environment-indicator vector $\mathbf{x}_i \in \{0,1\}^E$. In our main specification, $E = 22$ indicators represent 20 industries and 2 polarities. Each document belongs to one industry and one polarity. A concept token is one counted feature activation.

Each document is a mixture of K topics. The document-specific vector $\theta_i \in \mathbb{R}^K$ contains the topic logits, which determine the topic shares $\pi_i = \mathrm{softmax}(\theta_i)$. Topic k has global concept logits $\beta_k^\star \in \mathbb{R}^V$ and environment-specific deviations $\gamma_{e,k} \in \mathbb{R}^V$. For each document, its industry and polarity effects adjust these global logits.

We center the environment effects within each group g (industry or polarity). Let $u_{e,k}$ denote the uncentered effect of environment e on topic k. Define

$$\gamma_{e,k} = u_{e,k} - \sum_{f \in g} w_f^g u_{f,k}, \qquad e \in g,$$

where w_f^g is the proportion of documents used for estimation that belong to environment f. Thus, $\sum_{e \in g} w_e^g \gamma_{e,k} = 0$. We impose this restriction to distinguish the shared topic component from the environment effects.

Conditional on document lengths $N_i = \sum_{v=1}^V c_{i,v}$, the generative process is

1. For each topic $k = 1, \ldots, K$, draw $\beta_k^\star \sim \mathcal{N}(0, I)$ and uncentered environment effects $u_{e,k}$ from the shrinkage prior specified below. Center the environment effects within each group to obtain $\gamma_{e,k}$.
2. For each document $i = 1, \ldots, n$, draw $\theta_i \sim \mathcal{N}(0, I)$ and set $\pi_i = \mathrm{softmax}(\theta_i)$.
3. For each concept token $j = 1, \ldots, N_i$ in document i, draw a topic assignment and then a concept,

$$z_{ij} \sim \mathrm{Cat}(\pi_i), \qquad c_{ij} \sim \mathrm{Cat}\left[\mathrm{softmax}\left(\beta_{z_{ij}}^\star + \sum_{e=1}^E x_{i,e}\gamma_{e,z_{ij}}\right)\right].$$

Marginalizing over topic assignments gives the document-level concept probabilities

$$\eta_i = \sum_{k=1}^K \pi_{i,k}\, \mathrm{softmax}\left(\beta_k^\star + \sum_{e=1}^E x_{i,e}\gamma_{e,k}\right). \tag{1}$$

The count vector therefore follows

$$\mathbf{c}_i \mid N_i, \eta_i \sim \mathrm{Multinomial}(N_i, \eta_i),$$

with log-likelihood $\sum_{v=1}^V c_{i,v} \log \eta_{i,v}$ up to the multinomial coefficient.

We shrink the environment effects toward zero using the prior

$$\tau_{e,k,v} \sim \mathrm{Gamma}(a_\tau, b_\tau), \qquad u_{e,k,v} \mid \tau_{e,k,v} \sim \mathcal{N}(0, \tau_{e,k,v}^{-1}).$$

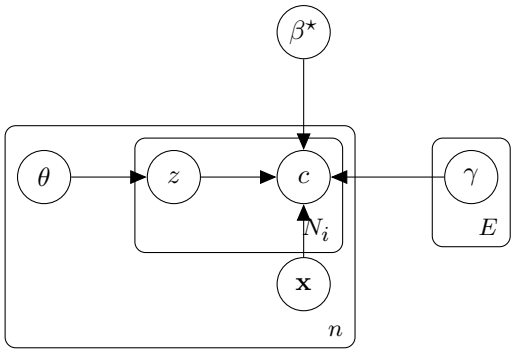


FIG 2. *Graphical model for mFETM. Document-specific logits θ determine topic assignments z. Each observed concept c depends on its assigned topic, global concept logits $\beta^\star$, environment deviations γ, and observed industry and polarity indicators $\mathbf{x}$. Plates index concept tokens within documents (N_i), documents (n), and environment levels (E).*

The shape and rate are fixed tuning parameters, reported in Supplementary Section C. We draw the precisions once and hold them fixed during estimation and evaluation. Centering $u_{e,k}$ gives the environment deviations $\gamma_{e,k}$.

The mVTM benchmark sets $\gamma \equiv 0$, so each topic has the same concept distribution across environments. Figure 2 shows the graphical model.

3.3. *Inference.* We condition on the fixed precisions τ and use variational inference to approximate the posterior over θ, $\beta^\star$, and u (Ranganath, Gerrish and Blei, 2014; Kingma and Welling, 2013; Blei, Kucukelbir and McAuliffe, 2017). We integrate out the topic assignments z and center each draw of u to obtain γ.

An encoder maps each document's concept counts to the mean and variance of a Gaussian approximation for θ_i. It has one hidden layer with 50 units, ReLU activation, and batch normalization. We maximize

$$\begin{aligned}\mathcal{L} = \mathbb{E}_q[\log p(\mathbf{c} \mid \theta, \beta^\star, \gamma(u))] - \sum_i \mathrm{KL}\{q_\phi(\theta_i) \,\|\, p(\theta_i)\} \\ - \mathrm{KL}\{q(\beta^\star) \,\|\, p(\beta^\star)\} - \mathrm{KL}\{q(u) \,\|\, p(u \mid \tau)\}\,,\end{aligned} \tag{2}$$

where KL denotes Kullback–Leibler divergence.

We estimate the variational means and variances of $\beta^\star$ and the means of u. The variances of u are fixed at τ^{-1}. After each Adam update, we center the variational mean of u (Kingma and Ba, 2014). We average the document log-likelihood and θ KL terms within each minibatch and divide the global $\beta^\star$ and u KL terms by the number of documents used for estimation.

Both models use the same data, representation, topic count, and encoder architecture. Optimization schedules are selected separately for each model. Both use the same validation-based checkpoint rule, and the test split is used only for final evaluation. Supplementary Section E reports the training and evaluation protocol.

3.4. *Baseline and Comparisons.* mVTM is the benchmark throughout. It uses the same representation, encoder architecture, and $K = 50$ topics, but sets $\gamma \equiv 0$. The comparison assesses what changes when observed environments enter the topic–concept distributions. We compare held-out fit, measure the association of topic shares with industry and review polarity, and test whether the shares distinguish firms included on Fortune's list.

TABLE 4
The leading concepts for an mFETM topic, its Con and Information-sector adjustments, and the matched mVTM topic.

	Top concepts
mVTM β	references to client experiences and services, terms related to information technology management, terms related to digital technology and information management, concepts related to multicultural strategies and organizational excellence
mFETM $\beta^\star$	phrases related to professional roles and responsibilities, concepts related to multicultural strategies and organizational excellence, references to strategies and roles related to corporate sustainability and management, key terms and phrases related to employment and job responsibilities
mFETM $\beta^\star + \gamma_{\mathrm{Con}}$	phrases related to professional roles and responsibilities, terms related to information technology management, references to strategies and roles related to corporate sustainability and management, concepts related to multicultural strategies and organizational excellence
mFETM $\beta^\star + \gamma_{\mathrm{NAICS}\,51}$	phrases related to professional roles and responsibilities, concepts related to multicultural strategies and organizational excellence, content related to business opportunities and strategies in sustainability and biotechnology, references to strategies and roles related to corporate sustainability and management

4. Topics and Predictive Fit. We estimate mFETM and mVTM with $K = 50$ topics on the same concept representation. Fixed effects allow mFETM to separate a shared theme from the language used to express it across environments. We illustrate this decomposition by comparing a matched topic across the two models, then examine how it affects held-out predictive fit.

4.1. *An Estimated Topic.* We illustrate the model outputs using the mFETM topic named "Enterprise Project Management and Integration." We compare its global component and two environment-specific adjustments with a matched mVTM topic. Supplementary Section B lists all topic names.

To name topic k, GPT-5.6 Luna receives 10 concept labels selected from the 12 concepts with the largest global weights and excerpts from the 20 reviews with the highest topic shares, evaluated at the posterior-mean logits. The review excerpts help interpret the concept labels in their workplace context. Naming takes place after estimation and uses neither environment labels nor environment-specific adjustments. Supplementary Section D gives the selection rule and prompt.

Table 4 presents the matched mVTM topic, mFETM's global logits $\beta^\star$, and two adjusted versions given by $\beta^\star + \gamma_{\mathrm{Con}}$ and $\beta^\star + \gamma_{\mathrm{NAICS}\,51}$. These show how the topic's concept distribution changes for cons and for the Information sector. Each adjustment adds one centered environment effect before applying the softmax.

The mVTM counterpart, "Enterprise Technology Strategy and Innovation," is selected by cosine similarity using cosine similarity between their centered concept-logit vectors. Its similarity to the mFETM topic is 0.79, compared with 0.59 and 0.57 for the next two matches.

The mVTM topic combines client services, information-technology management, digital information systems, and organizational strategy. In mFETM, professional roles and responsibilities is the leading concept in the global distribution and both environment-specific versions. Organizational strategy and corporate management also appear among the top four in all three. The environment effects change the emphasis within this shared theme. The Con effect raises information-technology management into the top four, while the Information-sector effect raises business strategies in sustainability and biotechnology.

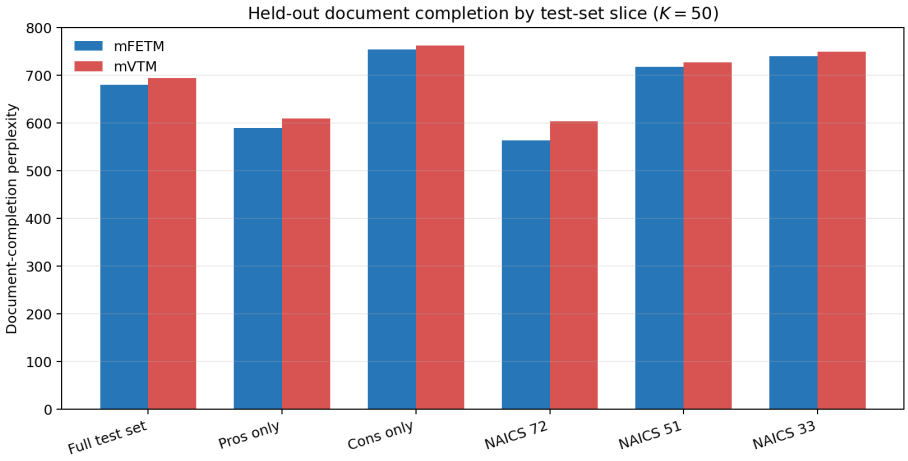


FIG 3. *Document-completion perplexity overall, by polarity, and for selected industries at $K = 50$. Lower values indicate better predictive fit. Both models predict the same held-out concept occurrences.*

mFETM thus represents the topic through a shared set of managerial concepts and explicit adjustments to their relative importance across environments. This gives the topic a common basis for comparison while retaining differences in how it is expressed. Supplementary Section F reports complementary geometric results. We next examine whether this separation comes at a cost to predictive fit.

4.2. *Predictive Fit: Perplexity.* We test whether separating shared themes from environment-specific language affects predictive fit. Both models use $K = 50$ topics, the same concept representation, and the same training, validation, and test sets. We evaluate document-completion perplexity on $90{,}348$ test documents. Perplexity measures how well a model predicts unobserved concept occurrences, with lower values indicating better predictions.

For each test document, we randomly divide its concept occurrences into two halves. We infer topic shares from one half and evaluate predictions for the other. Both models use the same split. mFETM achieves a perplexity of 680.5, compared with 694.6 for mVTM, a reduction of 2.0%. It also has lower perplexity for both polarities and all three industries shown in Figure 3.

mFETM uses each test document's observed industry and polarity to predict its concept counts. The improvement shows that separating shared and environment-specific content preserves information useful for prediction. We next test whether the topic shares are less associated with industry and polarity.

5. How Environment-Associated Are the Topics?. Topic shares intended to measure corporate culture may reflect contextual elements, complicating comparisons across firms. mFETM accounts for environment-specific language through fixed effects to make topic shares more comparable across environments. We assess this in two ways. First, we compare average topic shares across environments. Second, we test how accurately the full vector of topic shares predicts industry and review polarity.

5.1. *Topic-Share Dispersion.* For each topic k, we compute its average share $\bar{\pi}_{ke}$ in environment e and the spread $\Delta_k = \max_e \bar{\pi}_{ke} - \min_e \bar{\pi}_{ke}$. We calculate the measure across NAICS-2 industries with at least $1{,}000$ documents and across pro/con on the test split. A small spread indicates that a topic is used at similar rates across environments; a large spread may reflect either contextual language or genuine differences in theme prevalence.

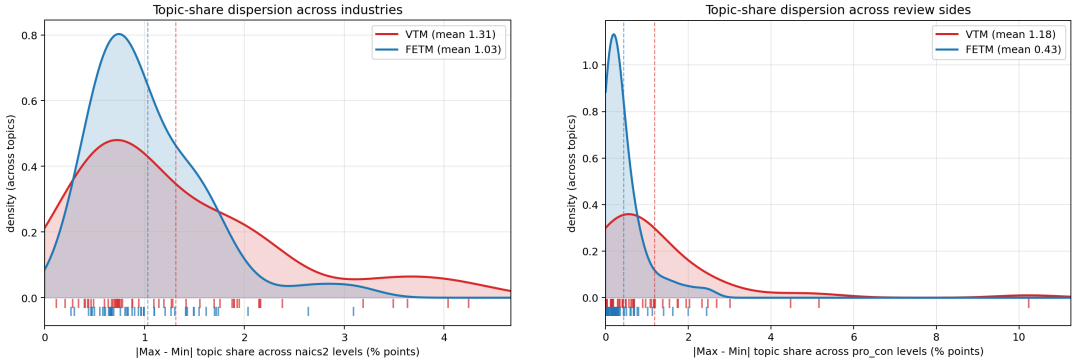


FIG 4. *Kernel density estimates of the per-topic spread Δ_k at $K = 50$. Left: industries, where the reduction under mFETM is not statistically distinguishable at the 5% level. Right: pro/con, where the reduction is larger and statistically significant.*

TABLE 5
Topic-share dispersion across environments (mFETM vs. mVTM, $K = 50$)

	Industries	Pro / Con
mVTM mean (% pts)	1.31	1.18
mFETM mean (% pts)	1.03	0.43
Difference (mFETM − mVTM)	-0.28*	-0.75***
	(0.16)	(0.25)
Welch t statistic	-1.72	-2.98
p-value (two-sided)	0.090	0.004
N (topics)	50	50

Notes. Each observation is a topic. "Max−Min" is the difference between the largest and smallest mean topic share across the levels of the column variable (NAICS-2 industries or Pro / Con). Topics are not aligned across models, so the test treats the two sets of topic-level values as independent samples (Welch two-sample t). Stars: $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$. Standard errors in parentheses. Held-out test split.

We compare the distributions of spreads across each model's 50 topics and test the difference in means using a two-sided Welch t-test. This comparison does not require matching topics across models. Replacing the maximum–minimum spread with the interquartile range also gives lower dispersion under mFETM.

Figure 4 shows the distributions across industries (left panel) and review polarity (right panel). Table 5 reports the tests. Across industries, the mean spread falls from 1.31 percentage points under mVTM to 1.03 under mFETM, a reduction of approximately 21%, statistically significant at the 10% level ($t = -1.72$, two-sided $p = 0.090$).

The reduction is larger across review polarity. The mean spread between pros and cons falls from 1.18 percentage points under mVTM to 0.43 under mFETM, a reduction of nearly two thirds ($t = -2.98$, two-sided $p = 0.004$). These results show that accounting for environment-specific language during estimation yields topic shares that are less tied to industry and, especially, review polarity.

5.2. *Predicting the Environment from Topics.* A topic may show little variation across environments on its own, yet reveal industry or polarity when combined with other topics. We therefore test whether the full vector of topic shares allows us to recover these environment labels. If mFETM separates environment-specific language from shared themes, its topic shares should reveal less about the environment than mVTM's.

TABLE 6
Predicting Pro / Con Review Side from Topic Shares ($K = 50$)

Contrast	N	mFETM R^2	mVTM R^2	mV − mF	mV / mF
Pro vs Con	906,938	0.197	0.554	0.356	2.8×

Notes. OLS of the pro/con indicator on an intercept and $K-1$ topic shares; the final share is omitted because the shares sum to one. The last two columns report mVTM minus mFETM and their ratio. Full corpus.

We first regress the pro/con indicator on the topic shares over the full corpus ($n = 906{,}938$),

$$\text{pro}_i = \alpha^m + \sum_{k=1}^{K-1} \delta_k^m \hat{\pi}_{ik}^m + \varepsilon_i^m, \qquad m \in \{\text{mFETM}, \text{mVTM}\}. \tag{3}$$

One topic share is omitted because the K shares sum to one.

Table 6 shows that mVTM's topic shares explain 55.4% of the variation in review polarity. Under mFETM, this falls to 19.7%, a reduction of approximately 64%. Accounting for polarity during topic estimation therefore substantially reduces the information about pros versus cons contained in the topic shares.

We next test whether topic shares distinguish industries. For each of the 28 pairs of eight selected two-digit NAICS sectors, we regress an indicator for which sector a document belongs to on its topic shares (Table 7). mVTM has the higher R^2 in every comparison. Its R^2 is 2.4 times mFETM's for Manufacturing versus Finance and 1.09 times mFETM's for Information versus Professional Services, the smallest difference. Together, these results show that mFETM's topic shares contain less information about both polarity and industry, even when all topics are considered jointly.

mFETM reduces differences in average topic shares across environments and makes industry and polarity harder to predict from the topics jointly. We next test whether these less environment-dependent measures retain the workplace information needed to distinguish top employers.

6. Topics and Top Employers. We next ask whether the topics capture workplace differences that distinguish top employers. Fortune-list inclusion provides an external benchmark of employer quality. We examine both how well each model explains inclusion and which workplace themes account for that fit. This allows us to assess whether mFETM retains substantive information about corporate culture while reducing its reliance on industry and review polarity.

We combine the Glassdoor reviews with Fortune's annual 100 Best Companies to Work For lists for 2009–2022. We identify candidate company matches using fuzzy matching and verify them manually. This links 76 Fortune companies to the Glassdoor panel, covering 425 firm-year appearances on the list.

We retain firm-years with at least five documents, yielding $7{,}992$ observations across 911 firms. Of these, 323 are listed firm-years. The remaining firm-years form the comparison group. We define $y_{f,t} = 1$ for listed firm-years and zero otherwise. Firms can contribute both listed and unlisted years.

For each model, we average document-level topic shares within each firm-year to obtain $\bar{\pi}_{f,t,k}^m$. Using $K = 50$ topics, we estimate

$$\Pr(y_{f,t} = 1) = \Lambda\left(\alpha_t^m + \sum_{k=1}^{K-1} \delta_k^m \bar{\pi}_{f,t,k}^m\right),$$

TABLE 7
Predicting Industry from Topic Shares ($K = 50$)

Industry contrast	N	mFETM R^2	mVTM R^2	mV − mF	mV / mF
Finance vs Accom & Food	178,456	0.210	0.283	0.073	1.3×
Accom & Food vs Health Care	86,546	0.094	0.159	0.066	1.7×
Retail vs Finance	245,677	0.125	0.185	0.060	1.5×
Information vs Accom & Food	221,425	0.227	0.286	0.058	1.3×
Accom & Food vs Prof Services	98,954	0.238	0.294	0.057	1.2×
Accom & Food vs Real Estate	90,384	0.135	0.191	0.056	1.4×
Information vs Retail	288,646	0.154	0.206	0.052	1.3×
Manufacturing vs Accom & Food	236,446	0.242	0.287	0.044	1.2×
Retail vs Health Care	153,767	0.037	0.078	0.041	2.1×
Manufacturing vs Finance	274,808	0.026	0.061	0.035	2.4×
Real Estate vs Health Care	36,836	0.040	0.074	0.035	1.9×
Manufacturing vs Retail	303,667	0.179	0.211	0.032	1.2×
Prof Services vs Health Care	45,406	0.096	0.126	0.030	1.3×
Manufacturing vs Health Care	182,898	0.041	0.069	0.029	1.7×
Retail vs Prof Services	166,175	0.099	0.125	0.026	1.3×
Information vs Health Care	167,877	0.039	0.060	0.021	1.5×
Finance vs Health Care	124,908	0.024	0.044	0.020	1.8×
Retail vs Real Estate	157,605	0.051	0.070	0.020	1.4×
Information vs Finance	259,787	0.020	0.037	0.017	1.9×
Retail vs Accom & Food	207,315	0.057	0.073	0.016	1.3×
Finance vs Real Estate	128,746	0.013	0.027	0.014	2.1×
Manufacturing vs Real Estate	186,736	0.021	0.033	0.011	1.5×
Manufacturing vs Information	317,777	0.017	0.029	0.011	1.6×
Information vs Real Estate	171,715	0.019	0.030	0.010	1.5×
Finance vs Prof Services	137,316	0.026	0.035	0.009	1.4×
Prof Services vs Real Estate	49,244	0.053	0.062	0.009	1.2×
Manufacturing vs Prof Services	195,306	0.012	0.021	0.009	1.7×
Information vs Prof Services	180,285	0.015	0.016	0.001	1.1×

Notes. Each row is a separate OLS of an industry indicator on an intercept and $K - 1$ topic shares. The final share is omitted because the shares sum to one. The last two columns report mVTM minus mFETM and their ratio. Full corpus.

where Λ is the logistic function and α_t^m are year fixed effects. Standard errors are clustered by firm. One topic is omitted because the shares sum to one.

We compare each topic's coefficient with the average coefficient of the other topics,

$$\Delta_k^m = \delta_k^m - \frac{1}{K-1}\sum_{j \neq k} \delta_j^m,$$

with the omitted topic's coefficient set to zero. A one-percentage-point increase in topic k's share, offset by equal reductions in the other shares, changes the log odds by $0.01\Delta_k^m$. Positive values indicate that shifting toward topic k is associated with a higher probability of Fortune-list inclusion. Negative values indicate a lower probability.

6.1. *Which Topics Are Associated with Fortune-List Inclusion?.* We examine which workplace themes distinguish firms on the Fortune list and whether accounting for industry and polarity makes these associations easier to interpret.

For each model, we test whether a greater share of each topic, relative to the other topics, is associated with Fortune-list inclusion. We adjust for testing 50 topics at once using the Benjamini–Hochberg procedure, applied separately to each model at a 5% false discovery rate. The main table reports the topics that remain statistically significant after this adjustment.

TABLE 8

Fortune top-100 topic predictors, mFETM ($K = 50$). Each coefficient contrasts one topic with the equally weighted average of all other topics and is therefore invariant to the fitting baseline. Rows pass within-model BH $q \leq 0.05$ across all 50 contrasts. The firm-year logit includes year FE and firm-clustered SEs. Stars use raw two-sided p-values: $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Topic (mFETM)	$\hat{\Delta}_k$ (SE)	BH q
26. Employee compensation and benefits	+39.89 (8.05)***	0.000
27. Company market positioning and competitiveness	+31.31 (7.22)***	0.000
10. Work-life balance and flexibility	+27.78 (6.53)***	0.000
8. Workplace Job Experience Evaluation	+14.06 (4.68)***	0.019
15. Employee perks and workplace amenities	+11.57 (4.26)***	0.037
24. Employee benefits packages	-14.40 (5.16)***	0.033
2. Abusive management and poor working conditions	-33.75 (10.29)***	0.009
37. Shareholder-driven restructuring and layoffs	-46.58 (8.41)***	0.000
35. Coworker friendliness and collegiality	-77.38 (14.87)***	0.000
N = 7,992; year FE; firm-clustered SEs; pseudo-R^2 = 0.1584; $\lvert z \rvert \geq 1.96$ and BH $q \leq 0.05$ topics only		

Table 8 identifies nine such topics under mFETM. Compensation and benefits, work-life balance, market positioning, general workplace experience, and amenities are positively associated with inclusion. Restructuring and layoffs, abusive management, benefit packages, and coworker friendliness are negatively associated with inclusion. These estimates compare each topic with the remaining topics. For example, the negative estimate for coworker friendliness means that greater emphasis on this theme relative to others is associated with lower inclusion, rather than that friendliness is undesirable.

Together, these topics describe concrete workplace dimensions spanning compensation, flexibility, coworker relations, management, job security, and amenities.

At the same threshold, mVTM identifies seven topics (Table 9). Food and beverage benefits, technology strategy, flexible work, and workload pressure are positively associated with Fortune-list inclusion. Abusive and retaliatory management, poor management, and workforce reductions are negatively associated.

mFETM provides a broader account of workplace differences and a better fit to Fortune-list inclusion. Alongside flexibility, benefits, management, and workforce reductions, it identifies compensation, market positioning, and general workplace experience. Its pseudo-R^2 reaches 0.1584, compared with 0.1363 for mVTM. Accounting for industry and polarity during topic estimation therefore strengthens the employer-quality associations while making the topics less dependent on those environments.

The source of that fit also matters. For instance, mVTM's significant predictors include technology strategy, a topic strongly associated with industry. Its fit may therefore partly reflect which sectors contain listed firms. We next remove the topics most associated with industry and review polarity to test whether mFETM's fit depends less on these environmental differences.

6.2. *Comparing mFETM and mVTM.* With all topics included, mFETM achieves a pseudo-R^2 of 0.158, compared with 0.136 for mVTM. We next test how much of this fit depends on topics associated with industry or review polarity.

Within each model, we rank topics by their association with these environments. We measure the standardized difference in average shares between pros and cons and the standardized range of average shares across industries. Each topic receives the larger of these two scores. The ranking does not use Fortune-list inclusion.

TABLE 9
Fortune top-100 topic predictors, mVTM ($K = 50$). The normalization-invariant contrast, selection rule, specification, and stars follow Table 8.

Topic (mVTM)	$\hat{\Delta}_k$ (SE)	BH q
5. Employee food and beverage benefits	+20.15 (5.35)***	0.003
6. Enterprise technology strategy and innovation	+16.73 (5.94)***	0.035
10. Flexible work arrangements and remote work	+15.25 (5.17)***	0.026
41. Workplace stress and workload pressure	+11.06 (3.75)***	0.026
28. Abusive and retaliatory management culture	-17.61 (5.93)***	0.026
15. Market downturns and workforce reductions	-26.54 (5.41)***	0.000
2. Poor management practices	-46.69 (10.12)***	0.000
N = 7,992; year FE; firm-clustered SEs; pseudo-R^2 = 0.1363; $\lvert z\rvert \geq 1.96$ and BH $q \leq 0.05$ topics only		

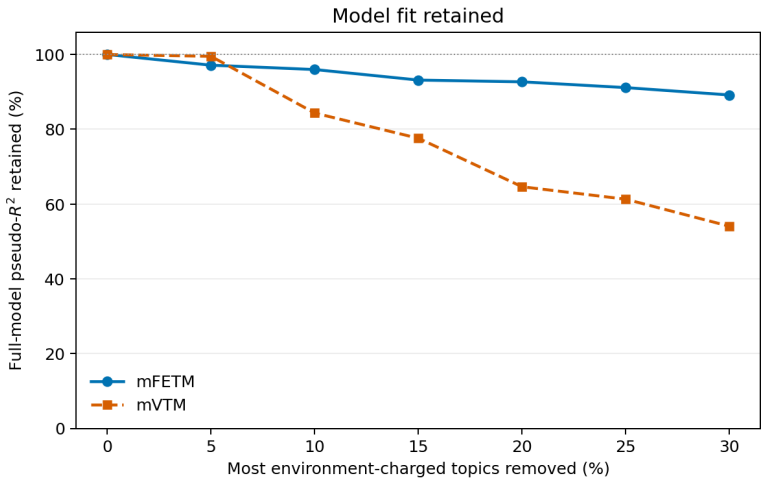


FIG 5. *Percentage of original fit retained after excluding the topics most associated with industry and review polarity. Fit is measured by McFadden pseudo-R^2. Both models use $K = 50$ topics.*

We progressively exclude the highest-ranked topics, removing the same number from each model. After each removal, we rescale the remaining firm-year topic shares to sum to one and re-estimate the Fortune regression on the same sample with year fixed effects. The topic models remain fixed throughout. The exercise tests how much information about top employers remains once the most environment-associated topics are excluded.

Figure 5 shows the percentage of each model's original fit retained. Removing five topics (10%) leaves 96% of mFETM's fit, compared with 84% for mVTM. Removing fifteen (30%) leaves 89% and 54%, respectively. Nearly half of mVTM's fit therefore disappears when its most environment-associated topics are removed, while mFETM retains almost nine tenths. mFETM both fits Fortune-list inclusion better and depends less on topics tied to industry and polarity.

Supplementary Section H reports the ranking and cutoff definitions, the Vuong comparison, a random-removal benchmark, and the results of removing topics based on polarity alone.

Nearly half of mVTM's fit disappears when its most environment-associated topics are removed, while mFETM retains almost nine tenths. Accounting for industry and polarity during estimation yields measures that distinguish top employers without relying as heavily on environmental differences.

7. Steering Along Estimated Fixed-Effect Directions. mFETM separates shared topics from environment-specific language. We now use the estimated environment effects to steer language generation. Holding the topic index and prompt fixed, we examine how generated text changes when we steer toward one industry or review polarity rather than another.

Each retained concept v has an SAE decoder direction $\mathbf{W}_{\text{dec}}[v] \in \mathbb{R}^{3584}$. For topic k in environment e, the fitted concept distribution is

$$p_{e,k} = \text{softmax}(\beta_k^\star + \gamma_{e,k}).$$

We map this distribution into the language model's activation space by taking a weighted average of the concept directions and normalizing its length,

$$\mathbf{s}_{e,k} = \frac{\sum_v p_{e,k,v} \mathbf{W}_{\text{dec}}[v]}{\|\sum_v p_{e,k,v} \mathbf{W}_{\text{dec}}[v]\|_2}.$$

To compare two environments a and b within the same group, we instead use the difference between their concept distributions,

$$\mathbf{s}_{a-b,k} = \frac{\sum_v (p_{a,k,v} - p_{b,k,v}) \mathbf{W}_{\text{dec}}[v]}{\|\sum_v (p_{a,k,v} - p_{b,k,v}) \mathbf{W}_{\text{dec}}[v]\|_2}.$$

The direction contrasts concepts more common under a with those more common under b. Its sign determines whether we steer toward a or b. Both distributions share the same topic backbone, and the other environment group is held at its centered mean.

Following Zheng et al. (2025), we intervene on Gemma's residual-stream activations at decoder blocks 40 and 41. Let $\mathbf{a}$ be the activation vector, $\mathbf{b}$ the SAE decoder bias, and $\mathbf{s}$ the selected unit direction. After subtracting the bias, we replace the component along $\mathbf{s}$ with $\lambda\mathbf{s}$ and leave the orthogonal component unchanged,

$$\bar{\mathbf{h}} = \mathbf{h} - \mathbf{b}_{\text{dec}}, \qquad \bar{\mathbf{h}}_\perp = \bar{\mathbf{h}} - (\bar{\mathbf{h}} \cdot \mathbf{s})\mathbf{s}, \qquad \mathbf{h}_{\text{steered}} = \bar{\mathbf{h}}_\perp + \lambda\mathbf{s} + \mathbf{b}_{\text{dec}}.$$

The parameter λ controls the strength and sign of the intervention. The language model's weights remain fixed. Within each comparison, we use the same topic, prompt, and generation procedure.

We illustrate the procedure using Topic 8, "Workplace Job Experience Evaluation." This broad theme permits comparisons across industries and review polarities. The same construction applies to any estimated topic.

We compare review polarity and industry expression within Topic 8. For polarity, we use opposite signs of the fitted Pro-minus-Con direction to steer toward pros or cons. We examine whether the continuations differ in how they evaluate workplace experience. For industry, we contrast Accommodation and Food Services (NAICS 72) with Information (NAICS 51) and examine differences in vocabulary and workplace setting.

All continuations begin with the same review prompt, reported in Table 10. We use steering strengths of $\lambda = +400$ and $\lambda = -400$ within each comparison to make the differences visible.

Table 10 shows how the estimated directions change the continuations. In Panel A, Pro steering emphasizes rewarding work, diversity, and employee commitment, while Con steering emphasizes difficulty and limited resources. In Panel B, steering toward Accommodation and Food Services produces references to customer service and a busy workplace, while steering toward Information produces a corporate-office setting. mVTM has no separately estimated environment effects from which to construct these directions.

We next assess the polarity response using held-out reviews. We select Topics 1–10 before inspecting the results and use the same ten Pro and ten Con documents for every topic and steering strength $\lambda \in \{-400, -300, \ldots, 400\}$. At each strength, we calculate the mean per-token log likelihood separately for Pro and Con reviews, take their difference, and subtract

TABLE 10
Generated continuations under polarity and industry steering for Topic 8.

Common prompt. "Fun atmosphere. Worked very independently. A lot of really great people at the company. The industry has its pros and cons."

Panel A. Review polarity	
Pro	
$\lambda = +400$	The job was rewarding because I got to work with a diverse group of employees from all over the world who were passionate about their jobs; however it also had some challenges such as long hours which made me feel like my life experience would be enriched by working there for an extended period time (which is why this position ended up being one that lasted 10 years).
Con	
$\lambda = -400$	The work was difficult but I learned a ton about how to get things done quickly with little resources (time/money). It's been hard for me because it requires more creativity than what my current job does which isn't as much creative thinking involved either way so that might be why this seems harder sometimes too?
Panel B. Industry	
Accommodation and Food Services	
NAICS 72, $\lambda = +400$	The customer service was good but not as fun to work with because they were a bit more stressful than some other customers I had worked with before during my time there (I am sure this is due mostly on how busy it gets). Overall though overall experience would be 5/10 if you're looking for something relaxing after school hours then definitely check out their website!
Information	
NAICS 51, $\lambda = -400$	The corporate office was a big plus for me as I am an out going person who likes to be around other companies/people all day long instead of sitting alone behind my desk every single minute!

Notes. The topic, prompt, and generation procedure are fixed across rows. Within each panel, the two continuations use opposite signs of the same estimated direction. Interventions are applied at decoder blocks 40 and 41.

its value at $\lambda = 0$. An increasing difference indicates that steering shifts relative likelihood toward Pro reviews as λ increases and toward Con reviews as it decreases.

Figure 6 shows that the average change rises almost linearly from -0.544 at $\lambda = -400$ to 0.514 at $\lambda = 400$. Nine of the ten topic directions produce the expected sign at both endpoints.

The continuations illustrate how the estimated effects alter language. The held-out likelihood results show a consistent polarity response across the selected topics. Together, they connect the statistical environment effects to changes in the language model's behavior. Supplementary Section G reports the accompanying fluency diagnostic.

8. Conclusion. Comparisons based on topic models require topics with a common meaning across documents. In multi-source corpora, differences in vocabulary and expression can obscure shared themes. Word counts also miss information carried by context and phrases. The mechanistic Fixed-Effects Topic Model addresses these problems by representing documents through interpretable concepts and separating shared topic content from environment-specific language.

Applied to Glassdoor reviews, mFETM improves held-out predictive fit and produces topic shares less associated with industry and review polarity. The variation in polarity explained by topic shares falls from 55.4% under mVTM to 19.7% under mFETM. Industry is also less predictable in all 28 pairwise sector comparisons. Accounting for environments during esti-

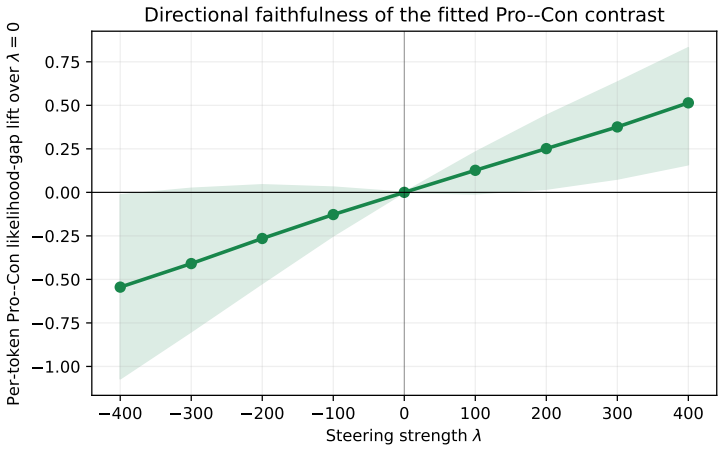


FIG 6. *Change in the Pro-minus-Con log-likelihood difference under steering. The line averages ten topic-specific directions, and the band shows their* 10*th–*90*th percentile range. Changes are measured relative to* $\lambda = 0$*, which removes the activation's projection onto each direction.*

mation thus reduces their influence on the resulting measures without sacrificing predictive fit.

These measures also distinguish top employers. mFETM identifies workplace dimensions spanning compensation, flexibility, management, job security, coworker relations, and amenities, while achieving a higher pseudo-R^2 for Fortune-list inclusion. Its fit depends less on environment-associated topics. Removing the most industry- and polarity-associated 30% of topics leaves 89% of mFETM's original fit, compared with 54% for mVTM. The information that distinguishes top employers therefore extends well beyond mFETM's most environment-associated topics.

The concept representation also allows us to use the estimated environment effects to steer the language model. Generated continuations illustrate changes in evaluative framing and workplace setting, while held-out likelihood comparisons show a consistent polarity response across the selected topics. The estimated effects can therefore be examined both as statistical parameters and through the language they produce.

mFETM offers a way to measure shared themes in corpora where sources differ systematically in how they express them. Its central contribution is to account for those differences when constructing the measures, giving researchers a common basis for comparison across environments.

SUPPLEMENTARY MATERIAL

Supplementary material for "The Mechanistic Fixed-Effects Topic Model"
Additional tables, topic names, concept-construction details, language-model prompts, inference algorithm, topic geometry, steering fluency diagnostics, and removal-sweep robustness checks.

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